

Pietro Sittoni

PhD Candidate, Applied Mathematics (GSSI)

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Summary

PhD candidate in Applied Mathematics at GSSI, advised by Francesco Tudisco (University of Edinburgh). I work on efficient machine learning for large-scale physics simulation and on architectural primitives inspired by mathematics, combining provable guarantees with large-scale experiments. Publications at ICML, ICLR and NeurIPS.

Experience

PhD Student – Machine Learning

Sep 2023 – Present

GSSI, L'Aquila, Italy

- Introduced **UFlex**, a minimal attention-based U-Net handling 1D/2D/3D spatiotemporal data with one model; scaled to 512×512 and $256 \times 128 \times 256$ grids on seven benchmarks, reaching state-of-the-art accuracy while cutting training memory and time versus transformer-based baselines. (**NeurIPS 2026**)
- Designed **Neural-HSS**, a parameter-efficient neural PDE solver built on Hierarchical Semi-Separable matrices and provably data-efficient; on 3D Poisson (2M grid points) it matches FNO trained on $4 \times$ more data and trains $2.4 \times$ faster. Validated on PDEs from electromagnetism, fluid dynamics and biology. (**ICML 2026**)
- Designed **SubDEQ**, a deep equilibrium model with existence and uniqueness guarantees under weaker assumptions than prior work, via nonlinear Perron–Frobenius theory. (**ICML 2024**)
- Built **Solaris**, a foundation model for the Sun: pre-trained a 3D Swin Transformer on 13 years of multi-wavelength imagery from NASA's Solar Dynamics Observatory for 12-hour-ahead forecasting, handling missing, corrupted and imbalanced samples; fine-tuned on an unseen wavelength in a low-data regime, it outperforms models trained from scratch. (**NeurIPS 2024 Workshop**)

Visiting Researcher

Oct 2024 – Dec 2024

University of Edinburgh, UK

- Co-developed **RecurrSim**, an architecture-agnostic framework for test-time control of the accuracy–cost trade-off in neural simulators; on 3D Navier–Stokes a 0.8B-parameter model beat 1.6B-parameter baselines with 13.5% less training memory, and RecurrViT cut error accumulation by 90% vs. standard ViTs. (**ICLR 2026**)

Research Intern

Dec 2022 – Sep 2023

GSSI, L'Aquila, Italy

- Extended nonlinear Perron–Frobenius theory to deep equilibrium models, laying the groundwork for SubDEQ.

Education

PhD in Applied Mathematics (Deep Learning)

2023 – 2027 (expected)

GSSI, L'Aquila, Italy. Advisor: Francesco Tudisco

MSc in Data Science – 110/110 cum laude, Università degli Studi di Padova, Italy

2021 – 2023

BSc in Mathematics, Università di Bologna, Italy

2017 – 2021

Publications

- Sittoni**, Tudisco. *Simplifying Transformer-Based U-Net Neural Physics Simulators*. NeurIPS 2026.
- Sittoni***, Zangrando*, Casulli, Guglielmi, Tudisco. *Neural-HSS: Hierarchical Semi-Separable Neural PDE Solver*. ICML 2026.
- Abdul Majid, **Sittoni**, Tudisco. *Test-Time Accuracy-Cost Control in Neural Simulators via Recurrent-Depth*. ICLR 2026.
- Sittoni**, Tudisco. *Algebraformer: A Neural Approach to Linear Systems*. NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences.
- Abdul Majid*, **Sittoni***, Tudisco. *Solaris: A Foundation Model of the Sun*. NeurIPS 2024 Workshop on Foundation Models for Science.
- Sittoni**, Tudisco. *Subhomogeneous Deep Equilibrium Models*. ICML 2024.

(*equal contribution)

Talks and Service

- Co-organizer, minisymposium *MS42: Artificial Intelligence and Scientific Computing*, SciCADE 2026, Edinburgh, UK.
- Reviewer for ICML, ICLR and NeurIPS.
- Talks: SciCADE 2026 (Edinburgh); Advanced Numerical Methods for Parametric PDEs (GSSI, 2026); Mathematics of Data-Driven Models (GSSI, 2026); ETNA Workshop (Catania, 2025).

Technical Skills

Python, PyTorch, TensorFlow, Hugging Face Diffusers, transformer architectures, generative models, NumPy, Pandas, Matplotlib/Seaborn, SLURM (HPC clusters), Git, YAML, $\LaTeX$.